

Tahsin Tajwar Tanni

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PERSONAL PROFILE

- Computer Science and Engineering graduate with a completed thesis on pretrained language model security, specialising in defence mechanisms against fine-tuning attacks and model robustness under adversarial conditions.
- Skilled in deep learning experimentation (PyTorch, Hugging Face Transformers), full-stack development (React, Node.js, MongoDB), and reproducible research workflows in Python

EDUCATION

01/2022 – 02/2026	<i>BRAC UNIVERSITY</i> BSc in Computer Science and Engineering	GPA: 3.60 / 4.00
03/2018 – 11/2020	<i>Govt. Haji Muhammad Mohsin College</i> Higher Secondary Certificate (HSC)	GPA: 5.00 / 5.00
01/2004 – 12/2018	<i>Bangladesh Mohila Samity Girls' High School and College</i> Secondary School Certificate (SSC)	GPA: 5.00 / 5.00

PROFESSIONAL EXPERIENCE

05/2025 – Present	AI Automation Trainee <i>180 RE, Toronto, Canada</i> • Built LLM-powered workflows using OpenAI and Anthropic APIs, automating multi-step business processes via n8n and third-party service integrations. • Developed and deployed AI-powered chatbots, integrating them into websites to automate user interactions and supports • Applied prompt engineering techniques to optimise LLM outputs for accuracy and safety across use cases.
02/2023 – 07/2024	Junior Web Developer <i>Weabers, Chattogram, Bangladesh</i> • Developed and maintained responsive web applications using modern frontend technologies. • Implemented reusable UI components, interactive features, and frontend logic aligned with project requirements. • Collaborated with backend developers to integrate APIs and dynamic data into frontend interfaces. • Debugged issues, refactored existing code, and assisted in deploying feature updates

RESEARCH EXPERIENCE

02/2025 - 07/2025	IT Instructor (Volunteer) <i>Zentorra</i> • Fundamental & practical concepts of Computer Science.
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RESEARCH EXPERIENCE

01/2025 – 01/2026 Undergraduate Researcher
BRAC University, Department of Computer Science and Engineering
Title: *Anchor-guided repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks*
(Accepted to IndabaX Nigeria 2026; forthcoming in *Proceedings of Machine Learning Research (PMLR)*, 2026).

- Large Language Models (LLMs) are vulnerable to post-release attacks such as weight noise injection and low-precision quantization, which degrade stability and performance. Users may unknowingly download compromised models without access to clean weights.
- We propose Anchor-Guided Repair, a defense that fine-tunes attacked models on clean text while limiting parameter changes via an anchor loss that penalizes deviation from a clean, task-adapted baseline.
- The method jointly optimizes language modeling loss and anchor regularization, reducing instability without distorting previously learned knowledge.
- Tested on various architectures and attack types (Gaussian noise, low-bit quantization), Anchor-Guided Repair consistently outperforms compromised models, restoring reliability close to the clean baseline.
- This approach demonstrates that anchoring provides a practical post-deployment defense, enabling safer and more trustworthy reuse of open-source LLMs without

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C

ML & AI: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas

Automation & MLOps: n8n, Weights & Biases, Docker, Hugging Face Spaces

Web & Databases: React.js, Node.js, Express.js, MongoDB, MySQL, SQLite

Tools: Git, GitHub, LaTeX

LANGUAGE PROFICIENCY

English: Full professional proficiency - Medium of Instruction (MOI) certified, BRAC University, 2026

Bengali: Native

AWARDS & HONORS

Dean's List, *BRAC University, Fall 2024*

Vice Chancellor's List, *BRAC University, Summer 2025*

2nd Runner-up – CSE471 Project Demonstration, *BRAC University, 2025*

Academic Distinction, BSc in Computer Science and Engineering, *BRAC University, 2026*

REFERENCES

Dr. Muhammad Iqbal Hossain
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Associate Professor, *BRAC University*
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Md. Aquib Azmain
Faculty Mentor
Senior Lecturer, *BRAC University*
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